

Permanently in Transient: Operationalizing Long-Memory Dynamics in the Autonomous-Vehicle Sensory Suite

Third companion to *Elliptic Analog Low-Pass Prototype: A Complete Zero–Pole–Gain Calculation* and *The Elliptic Prototype and the Kalman Filter: A Structural and Empirical Comparison*

“The filter is not merely sharp; it is, on any practical horizon, permanently in transient.”

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Chapter 1

From Pathology to Instrument

1.1 Provenance and assignment

The comparison report priced the $N = 293$ elliptic prototype's extremity and delivered the sentence quoted on the title page as an indictment: with its dominant pole at $\text{Re } p = -1.3827 \times 10^{-7}$, the filter's transients outlive any horizon on which a vehicle, a control loop, or an engineer operates. The present document executes a deliberate inversion of that verdict, prompted by the user-supplied conversion of the general transient-analysis taxonomy — the grammar of wellbore transient testing — into the autonomous-vehicle (AV) sensory suite. In the source discipline the transient is not a nuisance on the way to steady state; it *is* the measurement. A pressure record that has settled has stopped speaking. Under that reading, a system that never settles is a system that never stops integrating evidence, and “permanently in transient” names a resource: the long-memory end of a design dial that every estimator in the sensory suite already sits on, mostly by accident rather than by allocation.

This document makes the feature precise, derives its quantitative currencies, shows that the Kalman machinery of the comparison report computes the same dial from statistical inputs, and operationalizes it as concrete sensory-suite patterns expressed in the taxonomy's own grammar and three families (*reception validity*, *world-body characterization*, *action-envelope control*). Every quantitative claim is demonstrated numerically; the $N = 293$ prototype, verified to the last digit in the companion report, serves throughout as the calibrated extreme point of the dial.

1.2 Operational definition

Definition. A dynamical subsystem is *permanently in transient on horizon H* if its dominant time constant exceeds that horizon: $1/\varepsilon > H$, where ε is the distance of its slowest pole from the stability boundary (continuous axis or unit circle). On every horizon of action available to the stack, such a subsystem's state remains an unfinished integral of its input history.

Distinction. The feature is useful only when the memory is *addressable*: the long-lived state must be interpretable (a named physical or informational quantity), reportable with a calibrated uncertainty, and equipped with a reset policy. The $N = 293$ prototype's memory is real but incidental — a by-product of a magnitude template, storing an uninterpreted superposition in which a pothole rings for twenty hours alongside everything else. The patterns of Chapter 4 spend the same currency deliberately: narrowband resonators and low-noise Kalman trackers

whose permanent transient is pointed at a declared quantity. The exhibit demonstrates the currency; design spends it.

1.3 Position in the taxonomy

In the AV conversion's three families, this document contributes as follows. To *reception validity*: permanently-in-transient trackers are the natural custodians of synchronization integrity and estimator health, because drift and desynchronization are precisely the signals that only long memory can see. To *world-body characterization*: the perturbation and response primitives (micro-probes, pulses, step tests, passive imbalances) acquire a quantitative sizing rule — how small a probe can be, how long its answer takes, what resolution it buys. To *action-envelope control / refusal*: long memory is powerful and dangerous in equal measure, and Chapter 7 states the refusal rules — declared evidence vintage, mandatory fast companions, rebase hygiene, coherence matching — under which the feature may hold authority.

Chapter 2

The Currencies of Memory

2.1 One dial, four readings

Let a detector demodulate a probe line at frequency Ω_0 and integrate it with a leaky accumulator — the minimal permanently-in-transient element, a single pole at distance ε from the unit circle:

$$y_k = (1 - \varepsilon) y_{k-1} + x_k, \quad x_k = A e^{i\phi} + n_k, \quad \mathbb{E}|n_k|^2 = \sigma^2. \quad (2.1)$$

Its deflection signal-to-noise ratio after k samples is (Appendix A)

$$\text{SNR}(k) = \frac{A^2}{\sigma^2} \frac{(1 - \rho^k)(1 + \rho)}{(1 - \rho)(1 + \rho^k)}, \quad \rho = 1 - \varepsilon, \quad (2.2)$$

with two regimes that define the feature's economics:

$$\text{SNR}(k) \approx \begin{cases} \frac{A^2}{\sigma^2} k, & k \ll 1/\varepsilon \quad (\text{coherent growth, memory-independent}), \\ \frac{2A^2}{\varepsilon \sigma^2}, & k \gg 1/\varepsilon \quad (\text{saturation at the memory horizon}). \end{cases} \quad (2.3)$$

Every reading of the dial follows from (2.3). *Memory*: the integration horizon is $1/\varepsilon$ samples. *Sensitivity*: with a detection criterion $\text{SNR}^* = 13$ (deflection level corresponding to roughly $P_d \approx 0.9$ at $P_{fa} \approx 10^{-4}$), the smallest persistently detectable line is

$$A_{\min} = \sigma \sqrt{\text{SNR}^* \frac{1 - \rho}{1 + \rho}} \approx \sigma \sqrt{\frac{13}{2}} \varepsilon, \quad (2.4)$$

falling 10 dB for every 100-fold reduction in damping; the numerically fitted exponent of $A_{\min}$ against ε is 0.5017 against the theoretical $\frac{1}{2}$. *Resolution*: a detector of half-bandwidth ε separates two lines only if their spacing $\Delta\omega \gtrsim \varepsilon$ (Chapter 5 exhibits the transition). *Latency*: in the coherent regime, detection of amplitude A requires $k^* = \text{SNR}^* \sigma^2 / A^2$ samples — a hyperbola $A^2 k^* = \text{const}$ that is the same for every ε ; the memory setting decides only *where the hyperbola ends*.

2.2 The probe frontier

Figure 2.1 draws this frontier: probe amplitude against detection latency, one curve per memory setting, with Monte Carlo medians (40 runs per point) confirming the deflection theory —

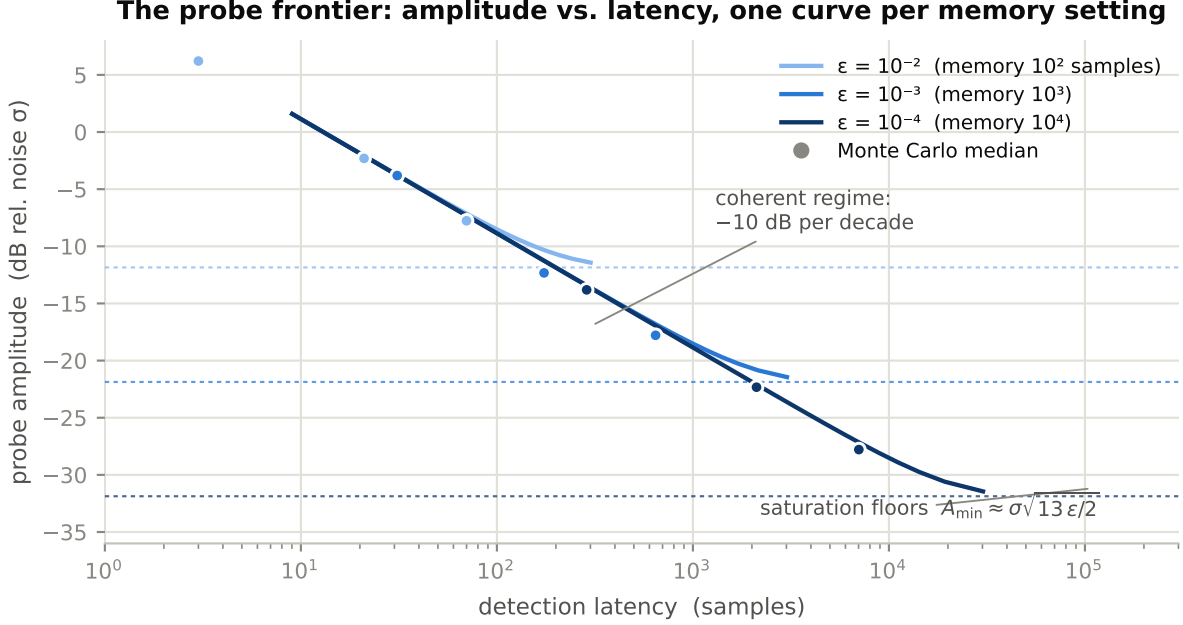


Figure 2.1: The probe frontier. In the coherent regime all memory settings share one -10 dB-per-decade trade; each setting departs the frontier at its own saturation floor $A_{\min} \approx \sigma\sqrt{13\varepsilon/2}$ (2.4). Dots: Monte Carlo medians.

measured latencies fall at 0.78–0.99 of the predicted values, the criterion being mildly conservative. The operational reading is the one the taxonomy’s perturbation rows need: *a persistent probe can be driven arbitrarily far below the noise floor, at a price paid in latency and memory*. A tire-torque dither at -32 dB relative to per-sample noise is invisible to any reflex-band detector and comfortably detectable at $\varepsilon = 10^{-4}$ within $\sim 10^4$ samples (Table 6.1 tabulates the full ladder).

2.3 Coherence matching

Memory is only worth buying while the world holds still. If the quantity carrying the probe’s answer (a friction transfer gain, a clock phase, a bearing line) wanders with coherence time T_c , integration beyond T_c averages incompatible worlds: the estimate’s variance stops falling and its *bias* begins to grow. The governing allocation rule is therefore

$$\varepsilon \gtrsim 1/T_c : \quad \text{memory no longer than the phenomenon’s coherence.} \quad (2.5)$$

Chapter 3 shows that the Kalman filter enforces (2.5) automatically — it is exactly what the Riccati equation computes from the model’s stated process noise.

2.4 The elliptic limit case, calibrated

Table 2.1 reads the four currencies at the far end of the dial, using the verified constellation of the companion documents. One number deserves emphasis: a hypothetical probe line integrated

Currency	Reading at the $N=293$ edge	Meaning
damping	$\varepsilon = 1.383 \times 10^{-7}$	distance of the dominant pole from the axis
memory	$1/\varepsilon = 7.232 \times 10^6$ samples	horizon over which history is retained
resolution	$\Delta\omega \approx \varepsilon = 1.38 \times 10^{-7}$	smallest separable frequency split
coherent gain	$10 \log_{10}(2/\varepsilon) = 71.6$ dB	SNR gain over a single sample
sensitivity	$A_{\min} = -60.5$ dB rel. σ	smallest detectable persistent line

Table 2.1: The four currencies read at the $N = 293$ band edge ($\text{SNR}^* = 13$ criterion for $A_{\min}$).

Instance	R_s (dB)	$\Delta\omega/\omega_p$	T_{pred}	$\tau(1)$ actual	ratio
$N = 293$ prototype	973.15	4.15×10^{-5}	1.03×10^7	7.37×10^6	1.40
$N = 6$ prototype	60	4.36×10^{-1}	4.35×10^1	2.19×10^1	1.99

Table 2.2: The delay–selectivity–order law (2.6) against the actual band-edge group delay of both verified prototypes.

at the $N = 293$ edge damping would be detectable at -60.5 dB relative to per-sample noise — 71.6 dB of coherent gain — if, and only if, the world agreed to hold its transfer gain coherent for 7.2×10^6 samples. The same structure explains the filter’s group delay through a compact heuristic: an order- N equiripple design distributing R_s decibels of discrimination across a relative transition $\Delta\omega/\omega_p$ must hold signal energy for

$$T \approx \frac{N 10^{R_s/(20N)}}{\Delta\omega/\omega_p}, \quad (2.6)$$

a delay–selectivity–order law verified within a factor of two on both elliptic instances of the comparison report — across nearly six decades of delay (Table 2.2, derivation in Appendix A). Selectivity in decibels is bought with memory, exponentially beyond what the order amortizes; “permanently in transient” is that purchase carried to 973 dB.

Chapter 3

The Kalman Route: λ as the Memory Dial

3.1 The oscillator-state tracker

The comparison report established the bridge in one direction: a steady-state Kalman filter is an LTI system with a zero–pole–gain triple. Here the bridge runs the other way: the natural *designed* carrier of a permanent transient is a Kalman tracker whose model asserts near-perfect coherence. Take the two-state oscillator model — the state rotates by Ω per sample, diffuses with process noise q per axis, and is observed in noise r :

$$F = \begin{bmatrix} \cos \Omega & -\sin \Omega \\ \sin \Omega & \cos \Omega \end{bmatrix}, \quad Q = qI_2, \quad H = [1 \ 0], \quad R = r, \quad \lambda \equiv q/r. \quad (3.1)$$

Solving the Riccati equation across ten decades of the tracking index λ and reading the closed-loop damping $\delta = 1 - |\text{eig}(I - \bar{K}H)F|$ gives a clean power law (Figure 3.1):

$$\delta \approx 0.70 \sqrt{\lambda}, \quad \text{fitted exponent } 0.4998. \quad (3.2)$$

The scalar local-level filter of the comparison report obeys the same law in closed form: its pole sits at $1 - \bar{K}$ with $\bar{K} = \frac{1}{2}(\lambda + \sqrt{\lambda^2 + 4\lambda}) \approx \sqrt{\lambda}$ (Appendix A). The interpretation is the central unification of this document: *the Kalman filter computes its own permanent transient*. The model’s stated incoherence rate λ — how much the world drifts per sample, in units of measurement noise — fixes the closed-loop damping at $\sqrt{\lambda}$, which is exactly the coherence-matching rule (2.5) executed automatically. Where Chapter 2’s resonator asks the designer to choose ϵ , the Kalman tracker derives it from a declared world model; the two dials are the same dial.

3.2 What the exhibit claims, in model language

Inverting (3.2) at the $N = 293$ edge damping $\delta = 1.3827 \times 10^{-7}$ gives the equivalent tracking index

$$\lambda_{\text{eq}} = 3.8 \times 10^{-14}. \quad (3.3)$$

A Kalman tracker becomes as long-memoried as the exhibit only when its model asserts that the tracked quantity drifts by $\sqrt{\lambda_{\text{eq}}} \approx 2 \times 10^{-7}$ measurement-standard-deviations per sample —

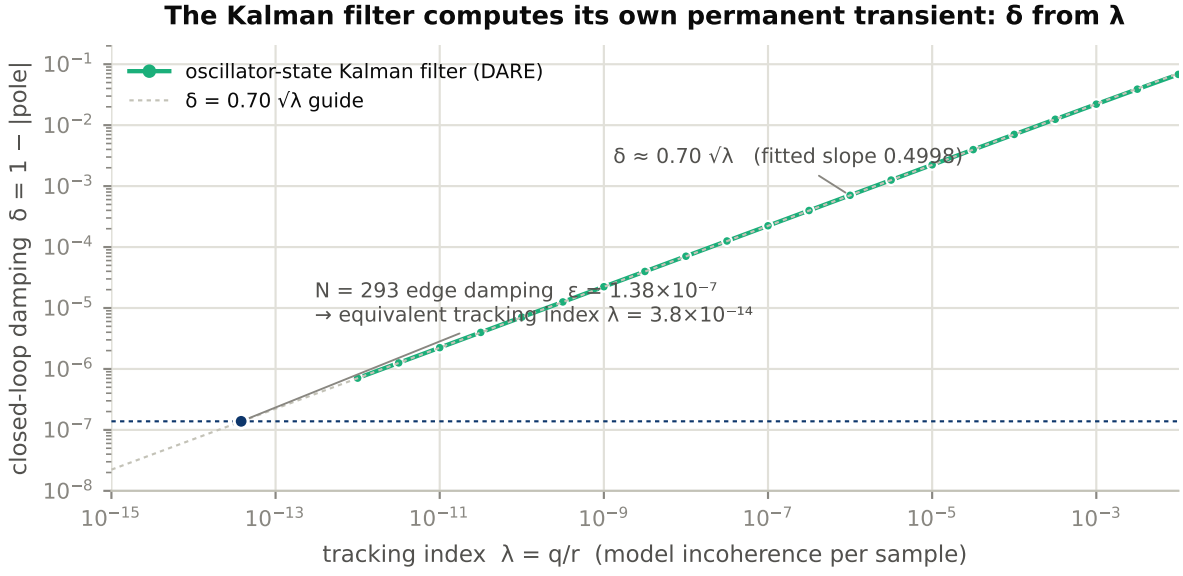


Figure 3.1: Closed-loop damping of the oscillator-state Kalman tracker versus tracking index (DARE solutions). The $N = 293$ edge damping corresponds to $\lambda = 3.8 \times 10^{-14}$.

roughly one part in five million. This is the statistical price tag of the elliptic specification, and it states the operational warning quantitatively: adopting a permanent transient is *making a coherence claim about the world*. Chapter 7 requires that claim to be explicit, monitored, and revocable.

3.3 Innovations: the settle-meter

The Kalman construction supplies one more instrument the resonator lacks. For a settled filter with a correct model, the innovation sequence $\nu_k = y_k - H\hat{x}_{k|k-1}$ is white with covariance S_k ; any *structured transient* in the innovations is precisely the signal that the model’s coherence claim has failed — a regime boundary in the taxonomy’s sense (“boundary detection,” “informational/learning-oriented signals”). This converts the document’s title from a property into a monitored contract: a permanently-in-transient estimator is licensed by the whiteness of its innovations, and loses its license — triggering the rebase policy of Chapter 5 — the moment its own surprise becomes structured.

Chapter 4

Operational Patterns for the Sensory Suite

The taxonomy’s grammar reads: *sensory-suit event = driving objective + world/body perturbation + multimodal response + source–receiver layout + latency class + sensor-contact geometry + control envelope + safety/inference model*. Each pattern below instantiates that sentence with the long-memory element as its inference model; Table 4.1 is the summary.

P1 — Sub-perceptual persistent probing (friction and actuator authority). Objective: transport-capacity estimation and maneuver-capacity assessment. Perturbation: “withdraw a small controlled quantity” — a persistent torque or brake dither at a declared line frequency, sized on the frontier of Figure 2.1. Response: wheel-speed, vibration, and IMU narrowband lines. Geometry: cross-segment (front-axle injection, wheel-end and body reception). Latency class: tactical ($\varepsilon \sim 10^{-3}$ to 10^{-4}). Envelope: probe amplitude below the passenger-perception and component-wear budget — the frontier shows -32 dB relative to per-sample noise remains detectable within $\sim 10^4$ samples at $\varepsilon = 10^{-4}$. Model: oscillator-state Kalman tracker at the dither line; the tracked complex gain *is* the tire–road transfer estimate, with calibrated variance. The taxonomy’s micro-perturbation archetype acquires, through (2.4), its missing sizing rule.

P2 — Long-memory condition monitoring (rotating-component lines). Objective: system-health inference (bearing, imbalance, driveline orders). Perturbation: none — opportunistic; the vehicle’s own rotation is the source. Response: acoustic/vibrational spectral lines at known orders. Latency class: surveillance ($\varepsilon \lesssim 10^{-6}$; memory of hours). A -52 dB line — far below anything a reflex- or tactical-band detector can see — is the natural signature of early mechanical degradation; only a permanently-in-transient tracker occupies that part of the frontier. Envelope: none required (passive). Model: resonator bank or oscillator trackers per order, rebased at maintenance events.

P3 — Synchronization-integrity phase ledger. Objective: reception validity — the taxonomy’s synchronization-integrity metric. Perturbation: none; the shared timeline is the phenomenon. Response: inter-sensor timestamp phase comparisons (camera–IMU–lidar–radar strobes). Model: near-unit-circle phase trackers (λ tiny because crystal drift is slow); their innovations, not their states, carry the alarm. Clock drift is the paradigm case of a quantity invisible on every short horizon and unmissable on a long one: the pattern’s entire value is its permanent transient.

P4 — Super-resolution line splitting. Objective: world–body characterization where two sources are nearly degenerate: left- versus right-side bearing orders, twin driveline harmonics,

Pattern	Taxonomy hooks	Perturbation	Latency band	Governing number
P1 micro-probe	micro-perturbation archetype; capacity assessment	persistent dither	tactical	$A_{\min} \approx \sigma \sqrt{13\varepsilon/2}$
P2 condition lines	acoustic/vibrational row; continuous surveillance	none (opportunistic)	surveillance	$-52 \text{ dB line at } \varepsilon = 10^{-6}$
P3 sync ledger	synchronization integrity	in- none (timeline)	surveillance	innovations, not states
P4 line splitting	scattering/EM rows; hidden-actor inference	optional	route/surv.	$\varepsilon \lesssim \Delta\omega, T \gtrsim 1/\Delta\omega$
P5 log-time probe	step-threshold test; boundary detection	step, held	reflex→thermal	humps at τ_i per tap
P6 innovation watch	informational signals; history inversion	none	all bands	whiteness license

Table 4.1: The pattern catalog, indexed to the AV taxonomy’s rows.

micro-Doppler gait lines of an occluded pedestrian against slow clutter. The two-tone law (Chapter 5) is the sizing rule: splitting $\Delta\omega$ requires $\varepsilon \lesssim \Delta\omega$ and observation beyond $1/\Delta\omega$; no short-memory detector, however good its SNR, can perform this discrimination.

P5 — Step-probe log-time diagnostics. Objective: operational-edge and stage attribution — *which* subsystem moved. Perturbation: a step (brake-pressure step, steering step) held under a constant-control condition. Response: the same probe read at multiple contact points (line pressure, pad force, slip, deceleration, temperature), each tap’s log-derivative completing on its own decade (Figure 5.3). This is the well-test derivative idiom imported whole: regimes appear as successive humps on logarithmic time, and a shifted hump names the shifted stage (pad fade versus tire-slip change versus thermal drift). Truncating the record — settling for the settled part — discards the still-arriving humps; the pattern is, by construction, an argument for remaining in transient as long as the safety case allows.

P6 — Innovation-transient watch. Objective: meta-safety — the taxonomy’s informational signal family (novelty, surprise, prediction error). Every settled estimator in the stack publishes innovations; a bank of cheap whiteness monitors watches them. The minimal instance is the fast/slow rebase trigger of Chapter 5 (fired 168 samples after a regime break); the general instance is fleet-level: recurring structured transients at the same road segment are a discovered hazard, the taxonomy’s “continuous operating-history inversion.”

Chapter 5

Numerical Demonstrations

5.1 Protocol

All demonstrations use unit-variance complex baseband noise, fixed seeds, and the deflection criterion $\text{SNR}^* = 13$; Monte Carlo latencies are medians over 40 runs. Table 5.1 collects every predicted-versus-measured pair; scripts and outputs accompany the document.

5.2 The frontier, measured

The Monte Carlo medians of Figure 2.1 track the deflection prediction at 0.78–0.99 across three decades of ε and two of amplitude; the fitted $A_{\min}$ exponent is 0.5017 against the theoretical $\frac{1}{2}$ (2.4). The frontier can be used as an engineering chart: pick the probe amplitude the comfort budget allows, read off the latency each memory band delivers, and refuse the inference if the required band exceeds the phenomenon’s coherence time (2.5).

5.3 Regime change: the price of memory, and its remedy

A channel gain drops by half mid-run. Detectors at $\varepsilon = 10^{-3}$ and 10^{-4} re-settle in 2,115 and 22,733 samples, against $\ln 10/\varepsilon = 2,303$ and 23,026 predicted — within 8% and 1.3%. (The $\varepsilon = 10^{-2}$ detector’s nominal 230-sample response is masked by its own noise floor: short memory pays in variance what it saves in agility, which is the frontier read backwards.) The remedy is architectural, not parametric: a fast companion ($\varepsilon = 10^{-2}$) plus a supervisory trigger — fire when the fast and deep estimates separate by more than four fast-band standard deviations, sustained — rebases the deep detector’s state onto the fast one. The trigger fired 168 samples after the break; the rebased deep detector recovered in 2,381 samples, $9.5\times$ faster than unaided, while retaining its -32 dB sensitivity floor (Figure 5.1). This fast/slow pairing is mandatory doctrine in Chapter 7.

5.4 Two-tone super-resolution

Two lines separated by $\Delta\omega = 2\pi \cdot 5 \times 10^{-7}$ per sample — a 5×10^{-7} relative split — are attacked by detectors at $\varepsilon = 10^{-3}, 10^{-5}, 2 \times 10^{-7}$ (Figure 5.2). Nothing separates before the decoherence horizon $1/\Delta\omega = 3.2 \times 10^5$ samples; the two shallow detectors never separate at all (measured

Demonstration	Predicted	Measured
$A_{\min}$ scaling exponent vs. ε	$1/2$	0.5017
Monte Carlo latency / deflection prediction	≤ 1	0.78–0.99
regime response, $\varepsilon = 10^{-3}$	$\ln 10/\varepsilon = 2303$	2,115
regime response, $\varepsilon = 10^{-4}$	23,026	22,733
rebased deep detector	—	2,381 (trigger at 168)
Kalman damping exponent vs. λ	$1/2$	0.4998
two-tone separation, $\varepsilon = 10^{-3}, 10^{-5}, 2 \times 10^{-7}$	0, ≈ 0 , 24 dB	0.00004, 0.41, 23.9 dB
delay law ratio ($N=293, N=6$)	≈ 1	1.40, 1.99

Table 5.1: Predicted versus measured, all demonstrations.

Deep memory reacts late — unless paired with a fast companion and a rebase rule

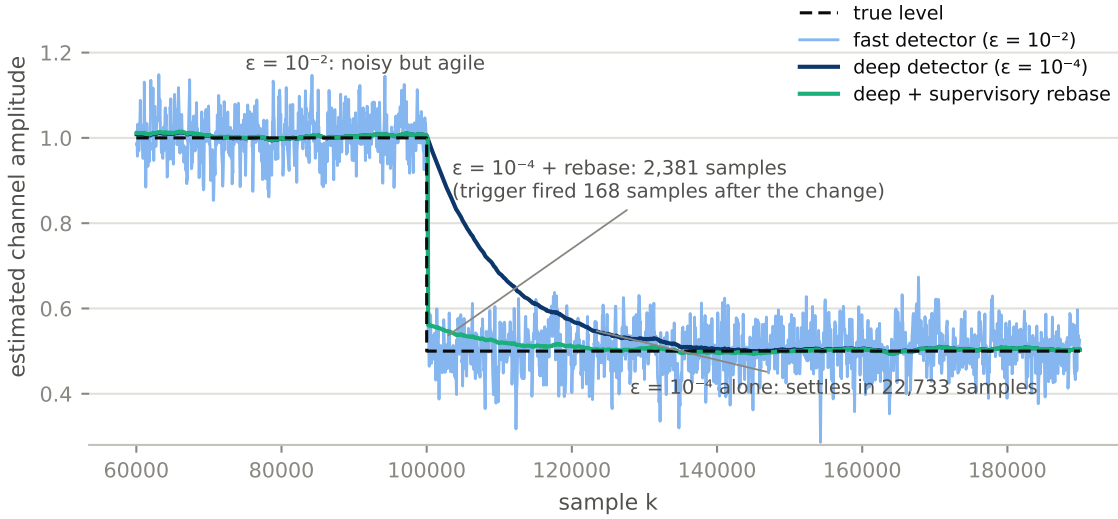


Figure 5.1: Memory reacts late unless supervised. The rebase rule restores agility without surrendering the deep detector’s sensitivity.

contrasts 0.00004 and 0.41 dB against their steady-state predictions); the deep detector, whose damping finally sits below $\Delta\omega$, completes its transient near $1/\varepsilon = 5 \times 10^6$ samples and delivers 23.9 dB of separation. Resolution is not an SNR problem — it is a memory problem, and only a permanently-in-transient element solves it.

5.5 The log-time diagnostic

Figure 5.3 executes pattern P5 on a five-stage brake-probe chain ($\tau = 0.02, 0.15, 0.8, 4, 90$ s: hydraulic, pad-rotor, tire slip, body dynamics, thermal). Read at five contact points, each tap’s log-derivative peaks on its own stage’s decade — measured peaks at 0.020, 0.152, 0.845, 4.30, and 89.8 s — reproducing the derivative-plot idiom of the source discipline: the record, viewed on logarithmic time, is a sequence of arriving regimes, and every truncation forfeits the regimes still en route.

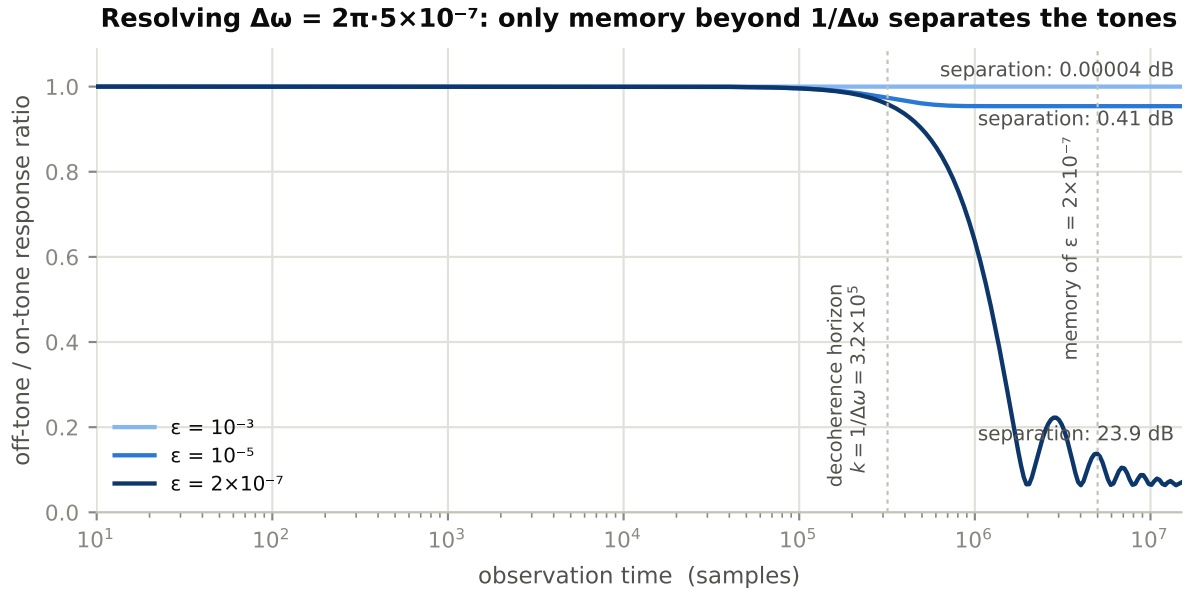


Figure 5.2: Splitting a 5×10^{-7} relative frequency pair: the discrimination arrives only after the decoherence horizon, and only for damping below the split.

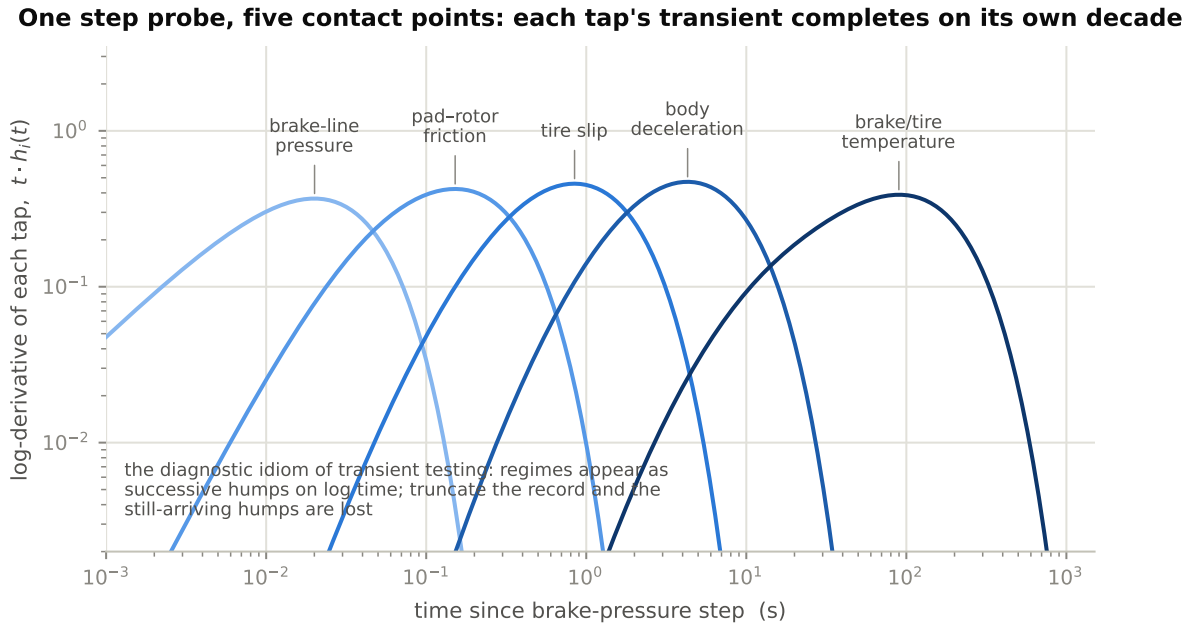


Figure 5.3: One step probe, five taps, five decades: the AV rendering of the transient-test derivative plot.

Chapter 6

The Transient Budget

The taxonomy’s five time-scale rows become five damping bands (Figure 6.1, Table 6.1): each estimator in the suite is *allocated* a memory horizon, rather than inheriting one from implementation accident. Three consequences follow. First, sensitivity is now a budgeted quantity: the table’s $A_{\min}$ column states what each band can and cannot see, so a requirement like “detect a -45 dB bearing line” is immediately a surveillance-band task with an hours-long evidence horizon — and cannot be levied on a tactical estimator. Second, agility is budgeted symmetrically: the regime-response column ($\ln 10/\varepsilon$) tells the planner how stale each band’s estimate can be after a boundary crossing, which is exactly the quantity the rebase doctrine bounds. Third, the $N = 293$ edge sits on the same chart at ≈ 20 hours of memory — beyond the surveillance band — as a calibrated reminder that the dial extends indefinitely and that every further decade of memory is a further coherence claim about the world.

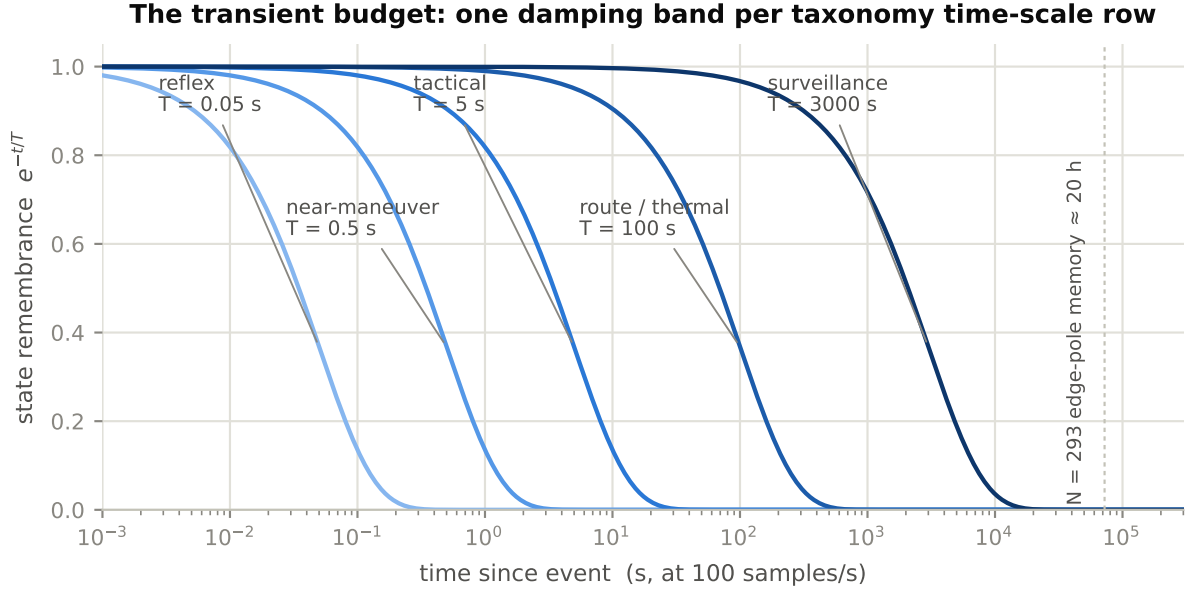


Figure 6.1: The transient budget at 100 samples/s: one damping band per taxonomy time-scale row, with the $N = 293$ edge as the calibrated extreme.

Band (taxonomy row)	T (s)	ε at 100 Hz	$A_{\min}$ (dB rel. σ)	regime response	authority
reflex	0.05	2.0×10^{-1}	1.1	0.12 s	full reflex actuation
near-maneuver	0.5	2.0×10^{-2}	-8.9	1.15 s	maneuver adjust/abort
tactical	5	2.0×10^{-3}	-18.9	11.51 s	maneuver selection
route / thermal	100	1.0×10^{-4}	-31.9	230 s	speed/route policy
surveillance	3000	3.3×10^{-6}	-46.6	6,908 s	veto and maintenance only

Table 6.1: Damping allocation across the taxonomy's time-scale rows (100 samples/s; $A_{\min}$ per (2.4); regime response $\ln 10/\varepsilon$).

Chapter 7

Envelope Control and Refusal

Long memory holds authority only under rules. These are stated in the taxonomy’s third-family voice: what the stack must publish, pair, bound, and refuse.

R1 — Declare the vintage. Every long-memory estimate is published with its horizon $1/\varepsilon$: the age of the evidence it embodies. A consumer may not use an estimate whose vintage exceeds the coherence time of the quantity it describes; the friction estimate of pattern P1 ages out at the weather’s time scale, not at the tracker’s.

R2 — Pair fast with deep. No permanently-in-transient estimator operates without a fast companion and a standing rebase rule (Chapter 5: trigger at 168 samples, recovery $9.5\times$ faster). The deep element supplies sensitivity; the fast element supplies the license to keep it.

R3 — Budget the probe. Persistent probes are sized on the frontier: at or above $A_{\min}(\varepsilon)$ for the allocated band, at or below the perceptual, wear, and energy budget. If the admissible amplitude window is empty, the probe — and the inference that wanted it — is refused, and the action envelope contracts accordingly.

R4 — Match memory to coherence. Rule (2.5): memory beyond the world’s coherence time is not information but bias. The Kalman formulation makes the claim explicit (λ is the declared incoherence), monitorable (innovation whiteness), and revocable (rebase). A resonator bank without an equivalent declared λ is not admissible in the suite.

R5 — Authority gradient. Authority horizon never exceeds memory honesty: reflex and near-maneuver authority rest only on reflex- and near-maneuver-band estimators. Surveillance-band estimators veto, schedule, and demand maintenance; they do not steer. The inverse also binds: a reflex layer may not overwrite (reset) a surveillance estimator except through the logged rebase mechanism of R6.

R6 — Reset hygiene. Rebase events are first-class data. Each one timestamps a detected regime boundary — the taxonomy’s boundary-detection output — and the fleet-level history of rebases is itself a surveillance signal (recurring boundary at one location = discovered hazard). Silent resets are prohibited: an estimator that forgets without a record has converted evidence into rumor.

Chapter 8

Validation

1. **Growth-and-saturation law.** Fitted $A_{\min}$ exponent 0.5017 against theoretical $\frac{1}{2}$; Monte Carlo detection latencies at 0.78–0.99 of deflection predictions across all nine spot checks (Table 5.1).
2. **Regime response.** Measured 2,115 and 22,733 samples against $\ln 10/\varepsilon$ predictions 2,303 and 23,026 (-8% , -1.3%); rebased recovery 2,381 samples with trigger at 168.
3. **Kalman dial.** DARE-fitted exponent 0.4998 against $\frac{1}{2}$; prefactor 0.70; local-level closed form consistent ($\bar{K} \approx \sqrt{\lambda}$); equivalent index of the $N = 293$ edge $\lambda_{\text{eq}} = 3.8 \times 10^{-14}$ read from the fitted law.
4. **Two-tone resolution.** Measured separations 0.00004, 0.41, 23.9 dB against steady-state predictions $\varepsilon/\sqrt{\varepsilon^2 + \Delta\omega^2}$ (0, ≈ 0.4 , 23.9 dB); separation onset at the predicted decoherence horizon 3.2×10^5 samples.
5. **Delay law.** Equation (2.6) within a factor 1.40 ($N=293$) and 1.99 ($N=6$) of the exact band-edge group delays computed from the verified constellations.
6. **Diagnostic peaks.** Five tap derivatives peak at 0.020/0.152/0.845/4.30/89.8 s against stage constants 0.02/0.15/0.8/4/90 s (Table 8.1).

Tap (contact point)	stage τ_i (s)	derivative peak (s)	window owner
brake-line pressure	0.02	0.0201	hydraulic subsystem
pad-rotor friction	0.15	0.152	brake pads/rotor
tire slip	0.8	0.845	tire-road interface
body deceleration	4	4.3	vehicle body dynamics
brake/tire temperature	90	89.8	thermal accumulation

Table 8.1: Log-time diagnostic: derivative-peak times against stage constants.

Appendix A

Derivations

Growth and saturation

With $y_k = \rho y_{k-1} + x_k$, $\rho = 1 - \varepsilon$, and $x_k = A + n_k$ after demodulation, the signal mean and noise variance after k steps are $\mu_k = A(1 - \rho^k)/(1 - \rho)$ and $v_k = \sigma^2(1 - \rho^{2k})/(1 - \rho^2)$. The deflection ratio μ_k^2/v_k gives (2.2); expanding $\rho^k = e^{k \ln(1-\varepsilon)} \approx e^{-\varepsilon k}$ yields the two regimes (2.3), and setting $k \rightarrow \infty$ with $\text{SNR} = \text{SNR}^*$ gives (2.4). The Monte Carlo detector declares at the first of three consecutive exceedances of $\text{SNR}^* v_k$ by $|y_k|^2$, which is mildly less conservative than the deflection criterion — hence measured latencies at 0.78–0.99 of predicted.

Local-level closed form

From the companion report, the steady-state predicted variance of the local-level model solves $\bar{P}^2 = Q(\bar{P} + R)$, so with $\lambda = Q/R$: $\bar{K} = \bar{P}/(\bar{P} + R) = \frac{1}{2}(\lambda + \sqrt{\lambda^2 + 4\lambda}) \rightarrow \sqrt{\lambda}$ as $\lambda \rightarrow 0$, and the closed-loop pole is $1 - \bar{K}$: damping $\delta = \bar{K} \approx \sqrt{\lambda}$. The oscillator-state model obeys the same square-root law with prefactor 0.70 (numerically, over $\lambda \in [10^{-10}, 10^{-4}]$).

The delay–selectivity–order heuristic

A single resonator of damping ε offers contrast $\sqrt{1 + (\Delta\omega/\varepsilon)^2} \approx \Delta\omega/\varepsilon$ at offset $\Delta\omega$, i.e., $D_1 = 20 \log_{10}(\Delta\omega/\varepsilon)$ decibels, so one section buying D_1 dB at offset $\Delta\omega$ must run at $\varepsilon = \Delta\omega 10^{-D_1/20}$ and hold memory $1/\varepsilon$. An order- N equiripple design spreads R_s decibels over N sections, $D_1 = R_s/N$ each; the edge sections dominate the delay, and summing N of them gives (2.6). The factor- ≤ 2 agreement in Table 2.2 reflects the crudeness of equal splitting; the exponential dependence on R_s/N — the economically decisive term — is exact in both instances.

Appendix B

Reproducibility

Computations in Python 3.11 (`numpy` 2.4, `scipy` 1.17, `matplotlib` 3.10). Scripts: `transient_demos.py` runs all demonstrations (seed 7, Monte Carlo 40 runs per point); `figures2.py` and `gen_tables3.py` generate every figure and table from the demonstrations' JSON outputs without hand transcription.

The $N = 293$ constellation values are those verified to 60 digits in the companion report; the taxonomy vocabulary follows the user-supplied autonomous-vehicle conversion.

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